

Assignment: Comparative Analysis of Machine Learning Models for Predictive Maintenance in Smart Manufacturing

A smart manufacturing company operates hundreds of CNC machines equipped with Industrial IoT (IIoT) sensors. These sensors continuously monitor vibration, temperature, motor current, rotational speed, and tool wear. The company wants to deploy a machine learning system that predicts whether a machine is likely to fail within the next 24 hours.

Machine failures can result in:

- Production downtime
- Financial losses
- Delayed customer deliveries
- Worker safety risks

Three machine learning models have been trained and evaluated using historical sensor data.

Model Performance Results

Model	Accuracy	Precision	Recall	F1-Score
SVM	90%	0.91	0.82	0.86
XGBoost	92%	0.89	0.90	0.89
Deep Neural Network (DNN)	94%	0.87	0.95	0.91

Additional Engineering Constraints

Model	Inference Time	Explainability	Deployment Cost
SVM	Low	High	Low
XGBoost	Medium	Medium	Medium
DNN	High	Low	High

Additional operational information:

- Missing an actual machine failure may cause approximately \$100,000 in production losses.
- A false alarm requiring unnecessary maintenance costs approximately \$5,000.
- Safety regulations require maintenance decisions to be explainable and auditable.
- Predictions must be generated in near real time.

Different stakeholders have different priorities:

- Maintenance engineers prioritize detecting failures as early as possible.
- Operations managers prioritize minimizing unnecessary maintenance interruptions.
- Financial managers prioritize cost reduction.
- Safety officers prioritize preventing catastrophic failures.

- Regulatory authorities prioritize explainability and accountability.

There is no single model that is best in every aspect.

Tasks

Task 1: Performance Metric Analysis

Explain the following metrics and their significance in predictive maintenance systems:

1. Accuracy
2. Precision
3. Recall
4. F1-Score

Additionally, discuss:

- Why accuracy alone may not be sufficient for evaluating predictive maintenance models.
- The importance of false positives and false negatives in this scenario.

Task 2: Comparative Analysis of Machine Learning Models

Analyze and compare the three machine learning models based on:

- Predictive performance
- Ability to detect machine failures
- Risk of false alarms
- Reliability in industrial environments

For each model:

- Identify its strengths.
- Identify its limitations.
- Discuss situations where the model may be preferred over the others.

Task 3: Engineering Decision-Making and Trade-Off Analysis

Assume you are the Lead AI Engineer responsible for selecting a model for deployment.

5. Recommend ONE model for deployment.
6. Identify at least three conflicting requirements that influence the decision.
7. Analyze the trade-offs among predictive performance, explainability, deployment cost, and real-time operation.
8. Discuss the potential consequences of choosing an unsuitable model.
9. Justify your recommendation using quantitative and qualitative evidence.

Note: There is no single correct answer. Marks will be awarded based on the quality of analysis, reasoning, and justification.

Task 4: Real-World Deployment Challenges

Discuss any three of the following challenges:

- Sensor noise and data quality issues
- Class imbalance in machine failure datasets
- Model interpretability and explainability
- Cybersecurity risks in Industrial IoT systems
- Scalability across multiple factories
- Continuous monitoring and model retraining
- Ethical and safety considerations
- Reliability of sensor infrastructure

For each challenge explain the issue, describe its impact on model performance, and suggest a mitigation strategy.

Task 5: Complex Engineering Problem Justification Report

Prepare a separate section titled "Justification of Complex Engineering Problem Attributes".

- P1 – Depth of Knowledge Required
- P3 – Wide-Ranging or Conflicting Requirements
- P7 – Significant Consequences in a Range of Contexts

Deliverables

Submit a report containing:

10. Performance Metric Analysis
11. Comparative Model Analysis
12. Engineering Decision and Trade-Off Analysis
13. Real-World Deployment Challenges
14. Complex Engineering Problem Justification Report
15. Conclusion

Marking Scheme (10 Marks)

Criterion	Marks
Performance Metric Analysis and Comparative Model Analysis	2.0
Engineering Decision & Trade-Off Analysis	2.0
Real-World Deployment Challenges	2.0
CEP Justification Report (P1, P3, P7)	2.0
Formatting of Report	2.0
Total	10